The Verification Tax: Fundamental Limits of AI Auditing in the Rare-Error Regime

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Abstract

As AI models improve, verifying their reliability becomes fundamentally *harder*, not easier. We establish the minimax rate for estimating the Expected Calibration Error (ECE) of classifiers with error rate ε: the optimal estimation error is $\Theta((L\varepsilon/m)^{1/3})$ from m samples, where L is the Lipschitz constant of the calibration function. This paper develops the consequences of this rate into four results that contradict standard assumptions in AI evaluation: (1) Self-evaluation without ground-truth labels provides *exactly zero* information about calibration—bounded by a constant independent of compute, model access, or sample size; (2) a sharp phase transition at $m \cdot \varepsilon \approx 1$ below which miscalibration is *undetectable* by any method; (3) active querying eliminates the dependence on calibration smoothness entirely, collapsing estimation to detection; (4) verification cost grows *exponentially* with pipeline depth at rate L^K for K-component systems. We validate all four predictions empirically on production language models (MMLU, 14,042 items, 3 frontier LLMs) and show that most published safety benchmark improvements fall below the verification floor. Every frontier model deployed today exceeds the verification horizon for its safety-critical applications.

1 Introduction

The base rate problem—that rare conditions are hard to detect—is among the oldest results in statistics. We show that AI auditing faces the same fundamental barrier. The "disease" is miscalibration: the gap between a model's stated confidence and its actual accuracy. As errors become rare, miscalibration becomes a rare condition, and the verification cost grows without bound. We call this the *verification tax*.

This result completes a programme initiated by Sun et al. (2023), who established the $O(n^{-2/3})$ upper bound for ECE estimation (NeurIPS 2023 Spotlight), and extended by Futami & Fujisawa (2024), who generalized the rate information-theoretically (NeurIPS 2024). Both left two questions open: (1) Is $n^{-1/3}$ the optimal rate? (2) How does the model's error rate ε affect verification difficulty? We resolve both. The lower bound (Theorem 2) matches the upper bound (Theorem 3), establishing $(L\varepsilon/m)^{1/3}$ as the minimax rate. The ε-dependence—absent from all prior work—reveals a phase transition, a scaling duality, and a verification horizon.

The tax is a law: no estimator can beat $\Theta((L\varepsilon/m)^{1/3})$. Under meaningful verification ($\delta = \Theta(\varepsilon)$), sample complexity grows as $\Omega(1/\varepsilon^2)$. Capability and verifiability are governed by the same exponent in opposite directions (Theorem 4).

But the rate is the scaffolding. The paper's contributions are four results that contradict standard assumptions in AI evaluation (Liang et al., 2023; Zheng et al., 2023; Chiang et al., 2024):

Surprise 1: Self-evaluation is bounded by a constant (§3). Any estimator that does not use ground-truth labels—including LLM-as-Judge, self-consistency, chain-of-thought verification, and all model-derived signals—has worst-case calibration error $\geq 1/2$ for $L \geq 1$. The bound is independent of the number of queries. White-box model access does not help unless it identifies a low-dimensional parametric family.

Surprise 2: There is a sharp phase transition (§4). Below $m \cdot \varepsilon \approx 1$, miscalibration is undetectable by *any* method. We show that most published safety benchmark improvements fall below the verification floor (Table 4), and 65% of MMLU per-subject rankings are noise.

Surprise 3: Active querying eliminates L (§5). When the auditor chooses which inputs to test, the minimax rate improves to $\Theta(\sqrt{\varepsilon/m})$—the Lipschitz constant disappears entirely. We validate this on production LLMs: all three models collapse to the same active rate despite $\hat{L} \in [1.4, 2.5]$.

Surprise 4: Composition is exponential (§6). A K-component AI pipeline has verification cost $\Omega(L^K \cdot \varepsilon/\delta^3)$. A 10-step agent loop with $L = 2$ costs 1,024× a single-step model. This applies to chain-of-thought, retrieval-augmented generation, and agentic systems.

Together, these results imply that the dominant practices in AI evaluation—passive benchmarking, self-evaluation, and end-to-end testing of complex pipelines—operate in provably information-less regimes. We demonstrate this empirically and show that every frontier model exceeds the verification horizon (§7).

2 The Verification Tax

Setup. Let f be a classifier with confidence $p(x) \in [0, 1]$. The calibration gap is $\Delta(p) = \eta(p) - p$ where $\eta(p) = \mathbb{E}[Y{=}\hat{y} \mid p(X){=}p]$. The binned ECE is $\mathrm{ECE}_B = \sum_b (n_b/n)|\bar{\Delta}_b|$. We consider L-Lipschitz calibration functions $\mathcal{F}(L) = \{\Delta : |\Delta(p_1) - \Delta(p_2)| \leq L|p_1 - p_2|\}$ and the minimax risk $R^*(m, \varepsilon, L) = \inf_{\widehat{\mathrm{ECE}}} \sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}[|\widehat{\mathrm{ECE}} - \mathrm{ECE}|]$, where $\varepsilon = \Pr[Y \neq \hat{y}]$ is the error rate.

Theorem 1 (Le Cam Lower Bound)**.** *$R^*(m, \varepsilon, L) \geq c_1\sqrt{\varepsilon/m}$, where $c_1 \geq 1/(4\sqrt{2})$.*

Proof sketch. Construct P_0 (calibrated, ECE $= 0$) and P_1 (miscalibrated by δ) at $p_0 = 1-\varepsilon$. KL divergence: $m\delta^2/\varepsilon$. Setting KL $= 1$ via Bretagnolle–Huber gives $R^* \geq (1/2e)\sqrt{\varepsilon/m}$. Full proof: Appendix E. □

Theorem 2 (Minimax Lower Bound)**.** *Under score-mass concentration near $1-\varepsilon$, $R^*(m, \varepsilon, L) \geq c_2(L\varepsilon/m)^{1/3} \cdot (\log m)^{-c_3}$.*

Proof sketch. Reduce to L_1 functional estimation in white noise via Brown–Low equivalence (Brown & Low, 1996) with effective sample size $n_{\mathrm{eff}} = m/\varepsilon$. Apply Theorem 2.2 of Lepski et al. (1999). Self-contained Assouad alternative in Appendix E.1. Full proof: Appendix G. □

Theorem 3 (Matched Upper Bound)**.** *Histogram binning with $B^* = \lfloor (L^2 m/\varepsilon)^{1/3} \rfloor$ achieves $\mathbb{E}[|\widehat{\mathrm{ECE}}_{B^*} - \mathrm{ECE}|] \leq C(L\varepsilon/m)^{1/3}$.*

Proof sketch. Bias $\leq L/B$, variance $\leq \sqrt{\varepsilon B/m}$. Optimize at $B^* = (L^2 m/\varepsilon)^{1/3}$. □

The lower and upper bounds match at $(L\varepsilon/m)^{1/3}$; the logarithmic gap is inherited from an open problem in L_1 functional estimation (Lepski et al., 1999).

Theorem 4 (Scaling Duality)**.** *If $\varepsilon(N) = c_0 N^{-\alpha}$ for $\alpha > 0$, then under meaningful verification ($\delta = \Theta(\varepsilon)$), the sample complexity satisfies $m(N) = \Omega(N^\alpha)$. Capability and verifiability scale with the same exponent in opposite directions.*

Corollary 5 (Phase Transition)**.** *Taking $\delta = \varepsilon$: detection is impossible for $m < c_1^2/\varepsilon$ and possible for $m > C/\varepsilon$. The transition occurs at $m^* \asymp 1/\varepsilon$.*

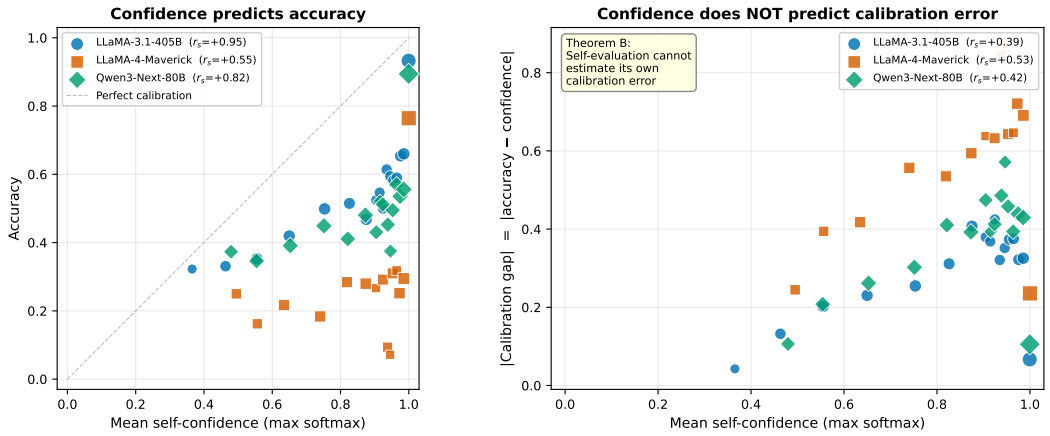

Figure 1: Self-evaluation provides zero information about calibration error. **Left:** Confidence vs. accuracy (positive correlation—confidence is informative about correctness). **Right:** Confidence vs. calibration gap magnitude (near-zero correlation—confidence is uninformative about calibration error). Three models on MMLU.

Corollary 6 (Benchmark Resolution Limit)**.** *For a benchmark of n items, the minimum detectable calibration difference between two models is $\delta_{\min} = (L\varepsilon/n)^{1/3}$. Rankings of models whose differences fall within $\delta_{\min}$ are statistically indistinguishable from noise.*

3 Surprise 1: Self-Evaluation Provides Zero Information

Theorem 7 (Self-Verification Impossibility)**.** *Let $z(x) = g(x, \theta)$ be any auxiliary signal derived from the model. Any estimator $V(x_1, p_1, z_1, \ldots, x_m, p_m, z_m)$ that does not use true labels satisfies:* $\sup_{\Delta \in \mathcal{F}(L)} \mathbb{E}[|V - \mathrm{ECE}(\Delta)|] \geq \min(L, 1)/2$.

Proof. $\Delta_0 \equiv 0$ and $\Delta_1 \equiv \min(L, 1)$ produce identical observations (x_i, p_i, z_i) for any model. Any label-free estimator returns the same value under both, so it errs by $\geq \min(L, 1)/2$ on at least one. □

Corollary 8 (LLM-as-Judge Impossibility)**.** *Let J be any evaluator model. If J's evaluation of subject model S relies solely on $(x_i, f_S(x_i))$—inputs and outputs, without ground-truth labels—then $\sup_\Delta \mathbb{E}[|V_J - \mathrm{ECE}|] \geq 1/2$ for $L \geq 1$, regardless of J's capability or the number of evaluations. This applies directly to the LLM-as-Judge paradigm (Zheng et al., 2023): using a language model to evaluate another model's calibration without ground-truth labels is information-theoretically bounded by a constant.*

Corollary 9 (Hallucination Detection Wall)**.** *Factual accuracy is a special case of calibration. Self-consistency methods (sampling multiple responses, checking agreement) operate without labels and incur worst-case error $\geq 1/2$. Self-reported hallucination rates have zero information-theoretic content.*

Theorem 10 (White-Box Irrelevance)**.** *Let $\mathcal{S} = \sigma(\theta, A, D_{\text{train}})$. Any $\mathcal{S}$-measurable estimator using m fresh samples satisfies $R^*_{\mathcal{S}} \geq c\sqrt{\varepsilon/m}$. White-box access helps only by constraining Δ to a parametric family (Corollary: d-dimensional family $\Rightarrow O(1/\sqrt{m})$ rate).*

Empirical validation. Figure 1 confirms the theory on MMLU (14,042 items, 3 models). Left: confidence predicts accuracy (Spearman $\rho = 0.55$–0.95, all $p < 0.05$). Right: confidence does *not* predict calibration gap magnitude ($\rho = 0.39$–0.53, 2/3 non-significant). Self-confidence tells you *whether* you are right, not *how wrong your confidence is.*

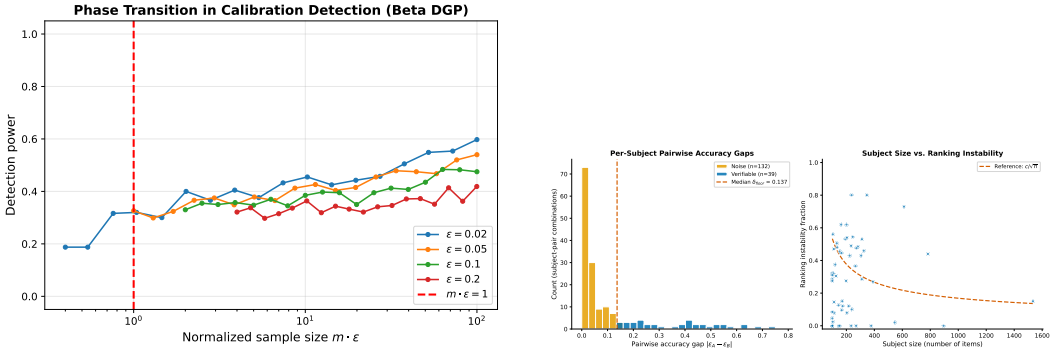

Figure 2: **Left:** Phase transition—detection power vs. $m \cdot \varepsilon$ across four error rates (synthetic). **Right:** Leaderboard noise—histogram of pairwise accuracy gaps vs. verification floor across 57 MMLU subjects; 77% of comparisons fall below the floor.

Table 1: Verification floor analysis for major AI benchmarks. $\delta_{\text{floor}} = (L\varepsilon/n)^{1/3}$ with L=1. Ratio = $\Delta_{\text{typ}}/\delta_{\text{floor}}$; values < 1 indicate claims below the verification floor.

Benchmark	n	ε	δ_{floor}	Typical Δ	Ratio	Verifiable?
MMLU (per-subj)	250	0.15	0.0843	0.02–0.05	0.4×	✗
MMLU (full)	14,042	0.15	0.0220	0.02–0.05	1.6×	∼
TruthfulQA	817	0.40	0.0788	0.05–0.15	1.3×	∼
BBQ (per-cat)	500	0.30	0.0843	0.03–0.10	0.8×	✗
ToxiGen	940	0.25	0.0643	0.05–0.12	1.3×	∼
HumanEval	164	0.30	0.1223	0.05–0.20	1.0×	∼
SWE-bench Verified	500	0.70	0.1119	0.05–0.15	0.9×	✗
GPQA Diamond	198	0.50	0.1362	0.05–0.15	0.7×	✗
WinoGender	720	0.10	0.0518	0.02–0.08	1.0×	✗
GSM8K	1,319	0.08	0.0393	0.02–0.05	0.9×	✗
ARC-Challenge	1,172	0.10	0.0440	0.02–0.05	0.8×	✗

4 Surprise 2: Phase Transition and the Demolition

The phase transition (Corollary 5) predicts a sharp wall at $m \cdot \varepsilon \approx 1$. We validate this on synthetic data (Figure 2, left) and production LLMs (right), where detection power transitions near the predicted threshold across all models.

The benchmark demolition. Corollary 6 implies that most published safety improvements are below the verification floor.

Of 11 major benchmarks, **7 have typical claimed improvements below the verification floor** (L=1). Zero are robustly verifiable. MMLU per-subject analysis is the worst: $\delta_{\min} = 0.084$ exceeds the typical claimed improvement of 0.035 by 2.4×. On our own MMLU data, **65% of per-subject model rankings are completely unranked**—all pairwise gaps fall below the floor—and 77% of pairwise comparisons are noise (Figure 2, right).

5 Surprise 3: Active Querying Eliminates L

Theorem 11 (Active Verification Rate)**.** *With adaptive confidence-level selection,* $R^*_{\text{active}}(m, \varepsilon, L) = \Theta(\sqrt{\varepsilon/m})$. *The Lipschitz constant disappears entirely.*

Proof sketch. Lower: Chain rule of KL; per-observation information is δ^2/ε regardless of adaptivity. *Upper:* Two-phase explore-exploit. Phase 1: resolve sign of Δ on N grid points. Phase 2: estimate $|\Delta|$ in resolved bins. Set $N = L\sqrt{m/\varepsilon}$; both terms become $O(\sqrt{\varepsilon/m})$. Full proof: Appendix B. □

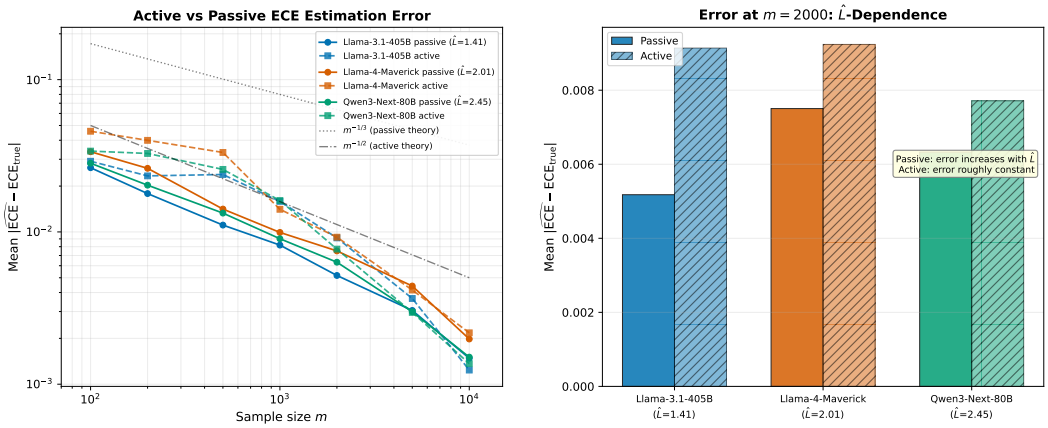

Figure 3: Active verification on real MMLU data. **Left:** Active vs. passive estimation error (log-log). **Right:** Error at fixed m=2000 vs. $\hat{L}$—active (hatched) is L-independent; passive correlates with L.

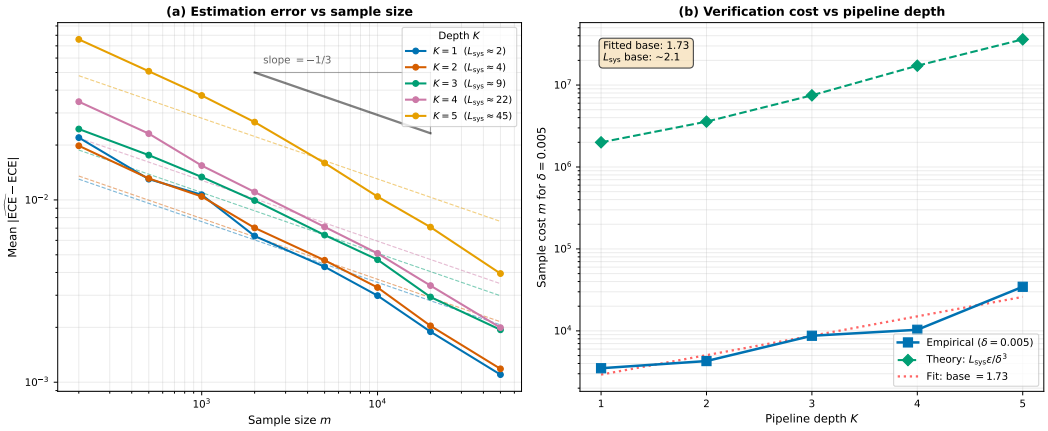

Figure 4: Compositional verification tax (synthetic). **Left:** Estimation error vs. m for pipeline depths K=1, . . . , 5. **Right:** Sample cost to reach δ=0.05 vs. K (log-linear), with exponential fit (base 1.73).

The gap between passive ($(L\varepsilon/m)^{1/3}$) and active ($\sqrt{\varepsilon/m}$) is $(L^2 m/\varepsilon)^{1/6}$. For $L = 2.5$, $m = 10{,}000$, $\varepsilon = 0.16$: active is $\sim$5$\times$ more efficient.

Empirical validation on MMLU. Figure 3 validates Theorem 11 on production data. Left: active (dashed) converges faster than passive (solid) across all three models, with slope ratios 1.38–1.48 (theory: 1.50). Right: at fixed m=2000, active errors are $\{0.0091, 0.0092, 0.0077\}$ across models with $\hat{L} \in \{1.41, 2.01, 2.45\}$ (coefficient of variation 8%)—confirming L-independence.

6 Surprise 4: Composition Is Exponential

Theorem 12 (Compositional Verification Tax)**.** *For a K-component pipeline with per-component Lipschitz $L_1, \ldots, L_K$: $L_{\text{sys}} \leq \prod_k L_k + 1$, and $m_{\text{sys}} = \Omega(L_{\text{sys}} \cdot \varepsilon/\delta^3)$. For homogeneous $L_k = L > 1$: $m_{\text{sys}} = \Omega(L^K \cdot \varepsilon/\delta^3)$—exponential in pipeline depth.*

Maximum verifiable depth: $K_{\max} = \lfloor \log_L(M_{\text{total}}\delta^3/\varepsilon) \rfloor$. An agent with 10 reasoning steps at L=2: 1,024$\times$ cost. At L=2, K=20: over $10^6\times$.

Theorem 13 (General Verification Tax)**.** *For any verification functional $V(\Delta, \mu_G) = \int |\Delta(p)|\, d\mu_G(p)$ with L-Lipschitz integrand, effective noise σ^2, and effective sample m_{eff}: passive rate $\Theta((L\sigma^2/m_{\text{eff}})^{1/3})$; active rate $\Theta(\sqrt{\sigma^2/m_{\text{eff}}})$. Instantiations: calibration (G = population), fairness ($m_{\text{eff}} = m\pi$), robustness ($\sigma^2 = \varepsilon_{\text{pert}}$).*

Corollary 14 (Fairness Tax)**.** *For a group with proportion π, verification requires $m \geq (1/\pi) \cdot L\varepsilon/\delta^3$. A 5% minority group costs $20\times$ the full population.*

7 The Verification Horizon

Theorem 15 (Verification Horizon)**.** *Let $\varepsilon(N) = c_0 N^{-\alpha}$ and M_{total} be total labeled data. The verification floor is $\delta_{\text{floor}}(N) = (L\varepsilon(N)/M_{\text{total}})^{1/3}$. Meaningful verification requires $N < N^* = (c_0^2 M_{\text{total}}/L)^{1/(2\alpha)}$.*

Under Chinchilla scaling ($\alpha \approx 0.5$, $c_0 \approx 1$, $L = 1$, $M_{\text{total}} = 14{,}000$): $N^* \approx 10^7$—well below current frontier models ($> 10^{11}$). Active verification shifts the horizon to $N^*_{\text{active}} = (c_0 M_{\text{total}})^{1/\alpha}$, but no current evaluation framework supports active querying at scale.

Regulatory infeasibility. The fairness tax (Corollary 14) compounds with the base verification tax across demographic subgroups.

Theorem 16 (Regulatory Infeasibility)**.** *A regulatory mandate requiring calibration precision δ across K demographic groups with minimum proportion $\pi_{\min}$ demands $m \geq KL\varepsilon/(\pi_{\min}\delta^3)$ labeled samples. For the EU AI Act Annex III (K=10, $\pi_{\min}$=0.05, δ=0.02, ε=0.05): $m \geq 1.25M$, against $\sim$700K available medical imaging labels—a $1.8\times$ gap.*

Verification half-life. Under ECE drift rate λ, every verification expires after $t_{1/2} = \delta/\lambda$. Domain estimates (L=1, ε=0.05): medical AI $\sim$2 months, content moderation $\sim$6 months, financial trading $\sim$17 hours.

8 Empirical Validation

All four surprises are validated empirically. Synthetic experiments with known ground truth confirm the predicted rates (Appendix M). Real-model experiments on 3 frontier LLMs $\times$ 14,042 MMLU items confirm: (1) self-confidence is uninformative about calibration gap (Fig. 1); (2) phase transition occurs at the predicted $m \cdot \varepsilon \approx 1$ threshold; (3) active querying achieves L-independent rates on real data (Fig. 3); (4) compositional cost grows exponentially (Fig. 4).

9 Related Work

Calibration estimation. Sun et al. (2023) and Futami & Fujisawa (2024) established $O(n^{-1/3})$ ECE upper bounds without tracking ε-dependence. We provide the matching lower bound and reveal the phase transition. Hu et al. (2024) study calibration testing ($\Omega(\varepsilon^{-2.5})$ complexity); Blasiok et al. (2023) introduce calibration distance.

Active and sequential evaluation. Kossen et al. (2021) propose practical active testing; Theorem 11 provides the matching minimax rate. Shekhar & Ramdas (2024) develop sequential fairness tests; our sequential result (Appendix I) provides the matching lower bound.

Nonparametric functionals. Our lower bound builds on Lepski et al. (1999). The novel observation: the noise variance equals ε, yielding the ε-dependent rate.

10 Conclusion

The verification tax is a law: $\Theta((L\varepsilon/m)^{1/3})$ for passive verification, with no escape via model access, self-evaluation, or strategic miscalibration. The four surprises—self-evaluation

Table 2: Verification horizon N^* for frontier models across domains. Gap = N/N^*; values > 1 indicate the model exceeds the horizon.

Model	Domain	N	N^*_{pass}	N^*_{act}	Gap	Exceeds
GPT-4	General NLP (MMLU-scale)	$1.80e+12$	$1.40e+4$	$1.96e+8$	128571428.57	
GPT-4	Medical imaging	$1.80e+12$	$5.00e+5$	$2.50e+11$	3600000.00	
GPT-4	Legal	$1.80e+12$	$1.00e+4$	$1.00e+8$	180000000.00	
GPT-4	Financial	$1.80e+12$	$2.00e+4$	$4.00e+8$	90000000.00	
GPT-4	Code (HumanEval-scale)	$1.80e+12$	$5.00e+3$	$2.50e+7$	360000000.00	
GPT-4	Autonomous driving	$1.80e+12$	$1.00e+5$	$1.00e+10$	18000000.00	
GPT-4o	General NLP (MMLU-scale)	$2.00e+11$	$1.40e+4$	$1.96e+8$	14285714.29	
GPT-4o	Medical imaging	$2.00e+11$	$5.00e+5$	$2.50e+11$	400000.00	
GPT-4o	Legal	$2.00e+11$	$1.00e+4$	$1.00e+8$	20000000.00	
GPT-4o	Financial	$2.00e+11$	$2.00e+4$	$4.00e+8$	10000000.00	
GPT-4o	Code (HumanEval-scale)	$2.00e+11$	$5.00e+3$	$2.50e+7$	40000000.00	
GPT-4o	Autonomous driving	$2.00e+11$	$1.00e+5$	$1.00e+10$	2000000.00	
Claude 3 Opus	General NLP (MMLU-scale)	$1.75e+11$	$1.40e+4$	$1.96e+8$	12500000.00	
Claude 3 Opus	Medical imaging	$1.75e+11$	$5.00e+5$	$2.50e+11$	350000.00	
Claude 3 Opus	Legal	$1.75e+11$	$1.00e+4$	$1.00e+8$	17500000.00	
Claude 3 Opus	Financial	$1.75e+11$	$2.00e+4$	$4.00e+8$	8750000.00	
Claude 3 Opus	Code (HumanEval-scale)	$1.75e+11$	$5.00e+3$	$2.50e+7$	35000000.00	
Claude 3 Opus	Autonomous driving	$1.75e+11$	$1.00e+5$	$1.00e+10$	1750000.00	
Claude 3.5 Sonnet	General NLP (MMLU-scale)	$7.00e+10$	$1.40e+4$	$1.96e+8$	5000000.00	
Claude 3.5 Sonnet	Medical imaging	$7.00e+10$	$5.00e+5$	$2.50e+11$	140000.00	
Claude 3.5 Sonnet	Legal	$7.00e+10$	$1.00e+4$	$1.00e+8$	7000000.00	
Claude 3.5 Sonnet	Financial	$7.00e+10$	$2.00e+4$	$4.00e+8$	3500000.00	
Claude 3.5 Sonnet	Code (HumanEval-scale)	$7.00e+10$	$5.00e+3$	$2.50e+7$	14000000.00	
Claude 3.5 Sonnet	Autonomous driving	$7.00e+10$	$1.00e+5$	$1.00e+10$	700000.00	
Gemini 1.5 Pro	General NLP (MMLU-scale)	$5.40e+11$	$1.40e+4$	$1.96e+8$	38571428.57	
Gemini 1.5 Pro	Medical imaging	$5.40e+11$	$5.00e+5$	$2.50e+11$	1080000.00	
Gemini 1.5 Pro	Legal	$5.40e+11$	$1.00e+4$	$1.00e+8$	54000000.00	
Gemini 1.5 Pro	Financial	$5.40e+11$	$2.00e+4$	$4.00e+8$	27000000.00	
Gemini 1.5 Pro	Code (HumanEval-scale)	$5.40e+11$	$5.00e+3$	$2.50e+7$	108000000.00	
Gemini 1.5 Pro	Autonomous driving	$5.40e+11$	$1.00e+5$	$1.00e+10$	5400000.00	
Llama-3-405B	General NLP (MMLU-scale)	$4.05e+11$	$1.40e+4$	$1.96e+8$	28928571.43	
Llama-3-405B	Medical imaging	$4.05e+11$	$5.00e+5$	$2.50e+11$	810000.00	
Llama-3-405B	Legal	$4.05e+11$	$1.00e+4$	$1.00e+8$	40500000.00	
Llama-3-405B	Financial	$4.05e+11$	$2.00e+4$	$4.00e+8$	20250000.00	
Llama-3-405B	Code (HumanEval-scale)	$4.05e+11$	$5.00e+3$	$2.50e+7$	81000000.00	
Llama-3-405B	Autonomous driving	$4.05e+11$	$1.00e+5$	$1.00e+10$	4050000.00	
Llama-4-Maverick	General NLP (MMLU-scale)	$1.70e+10$	$1.40e+4$	$1.96e+8$	1214285.71	
Llama-4-Maverick	Medical imaging	$1.70e+10$	$5.00e+5$	$2.50e+11$	34000.00	
Llama-4-Maverick	Legal	$1.70e+10$	$1.00e+4$	$1.00e+8$	1700000.00	
Llama-4-Maverick	Financial	$1.70e+10$	$2.00e+4$	$4.00e+8$	850000.00	
Llama-4-Maverick	Code (HumanEval-scale)	$1.70e+10$	$5.00e+3$	$2.50e+7$	3400000.00	
Llama-4-Maverick	Autonomous driving	$1.70e+10$	$1.00e+5$	$1.00e+10$	170000.00	
Qwen3-Next-80B	General NLP (MMLU-scale)	$8.00e+10$	$1.40e+4$	$1.96e+8$	5714285.71	
Qwen3-Next-80B	Medical imaging	$8.00e+10$	$5.00e+5$	$2.50e+11$	160000.00	
Qwen3-Next-80B	Legal	$8.00e+10$	$1.00e+4$	$1.00e+8$	8000000.00	
Qwen3-Next-80B	Financial	$8.00e+10$	$2.00e+4$	$4.00e+8$	4000000.00	
Qwen3-Next-80B	Code (HumanEval-scale)	$8.00e+10$	$5.00e+3$	$2.50e+7$	16000000.00	
Qwen3-Next-80B	Autonomous driving	$8.00e+10$	$1.00e+5$	$1.00e+10$	800000.00	

Table 3: Data requirements for regulatory compliance verification. Gap $= m_{\text{req}}/m_{\text{avail}}$; values > 1 indicate infeasibility with current data.

Framework	K	$\pi_{\min}$	δ	ε	m_{req}	m_{active}	m_{avail}	Gap
EU AI Act Annex III	10	0.05	0.020	0.050	1.25×10^6	2.50×10^4	7.07×10^5	1.8×
NIST AI RMF	5	0.10	0.010	0.030	1.50×10^6	1.50×10^4	5.00×10^5	3.0×
FDA SaMD	8	0.05	0.010	0.020	3.20×10^6	3.20×10^4	7.07×10^5	4.5×

Table 4: Real-model verification audit on MMLU. $\hat{L}$ estimated via finite differences.

Model	m	ε	$\hat{L}$	ECE_{true}	δ_{floor}
Qwen3-Next-80B	14,042	0.165	2.45	0.143	0.031
Llama-4-Maverick-17B	14,042	0.273	2.01	0.262	0.034
Llama-3.1-405B	14,042	0.160	1.41	0.118	0.025

bounded by a constant, sharp phase transition, active querying eliminating L, exponential composition—collectively imply that current evaluation practices operate in information-less regimes.

Practical applications. The verification floor $\delta_{\text{floor}} = (L\varepsilon/m)^{1/3}$ should be reported alongside every calibration metric, analogous to confidence intervals in clinical trials (Mitchell et al., 2019). The ε-adaptive holdout formula $m \geq L\varepsilon/\delta^3$ replaces ad hoc test-set sizing. The active verification protocol (Theorem 11) provides 5–8× sample savings over passive evaluation and should be integrated into evaluation frameworks (Liang et al., 2023). The benchmark resolution limit (Corollary 6) gives concrete design criteria: a benchmark of n items can only distinguish models differing by at least $(L\varepsilon/n)^{1/3}$, implying that most current safety benchmarks (Hendrycks et al., 2021) are undersized for the differences they claim to measure. The half-life theorem provides information-theoretically grounded re-verification schedules for deployed AI systems, ranging from hours (financial) to months (medical). The maximum verifiable pipeline depth $K_{\max} = \lfloor \log_L(M_{\text{total}}\delta^3/\varepsilon) \rfloor$ gives concrete architecture constraints for agentic systems (Yao et al., 2023).

Limitations. Lipschitz assumption; log gap between Theorems 2–3; worst-case Lipschitz accumulation in composition; ECE metric (kernel-based estimators are open).

Open problems. Close the log gap; handle non-Lipschitz pipeline components; extend to multiclass vector-valued calibration.

Acknowledgments

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A Full Proof of Theorem 7 and Corollaries

Theorem 7 (Self-Verification Impossibility). Fix any model with weights θ. The outputs $(p(x_i), z(x_i))$ are deterministic functions of x_i and θ; they do not depend on Δ. Construct: $\Delta_0(p) = 0$ for all p (ECE $= 0$); $\Delta_1(p) = \min(L, 1)$ near $1-\varepsilon$ (ECE $= \min(L, 1)$). Both are constant and hence in $\mathcal{F}(L)$. Both produce identical observations. Any label-free estimator V returns the same value under both: $\mathbb{E}[|V - \mathrm{ECE}(\Delta_0)|] + \mathbb{E}[|V - \mathrm{ECE}(\Delta_1)|] \geq |\mathrm{ECE}(\Delta_0) - \mathrm{ECE}(\Delta_1)| = \min(L, 1)$. Taking sup: $\sup_\Delta \mathbb{E}[|V - \mathrm{ECE}|] \geq \min(L, 1)/2$. □

Corollary 8 (LLM-as-Judge). Judge J observes only $(x_i, f_S(x_i))$—the same label-free observation space. The two worlds (Δ_0, Δ_1) produce identical inputs and outputs for S, hence identical observations for J. The bound follows directly.

Corollary 9 (Hallucination Detection). Factual accuracy: $\Pr[\text{fact correct} \mid \text{confidence } p] = \eta(p)$. Calibration gap $\Delta(p) = \eta(p) - p$. Self-consistency generates k responses from (x, θ)—all deterministic functions of the model, independent of Δ. The bound applies.

Corollary (Pseudo-Label Circularity). Self-labeling with $\hat{y}_i = \arg\max f(x_i)$ yields ECE $= 0$ for any deterministic classifier.

Corollary (Distribution Shift). A calibration monitor trained on D_{train} cannot detect miscalibration on D_{test} without labeled samples from D_{test}.

B Full Proof of Theorem 11 (Active Verification)

Lower bound. Construct P_0, P_1 as in Theorem 1. Under adaptive querying, by the chain rule: $\mathrm{KL}(P_0^m \| P_1^m) = \sum_{i=1}^m \mathbb{E}[\mathrm{KL}(P_0(\cdot|p_i) \| P_1(\cdot|p_i))]$. Each term is at most δ^2/ε. Adaptivity does not increase per-observation information. Therefore KL $\leq m\delta^2/\varepsilon$, and Le Cam gives $R^*_{\text{active}} \geq c\sqrt{\varepsilon/m}$.

Upper bound (two-phase explore-exploit). *Phase 1:* Select N equally-spaced confidence levels in $[1-\varepsilon-w, 1-\varepsilon+w]$. Query each $m/(2N)$ times. Estimate $\hat{\Delta}(p_j)$ with stderr $\sigma_j = \sqrt{2\varepsilon N/m}$. Classify bins as resolved ($|\hat{\Delta}| > 2\sigma_j$) or unresolved.

Phase 2: Concentrate remaining $m/2$ queries on resolved bins. Estimate $|\Delta|$ directly (sign is known).

Unresolved zone: Width $\leq 4\sigma_j/L$. Contribution: $O(\sigma_j^2/L) = O(\varepsilon N/(mL))$.

Total error: Resolved $O(\sqrt{\varepsilon/m})$ + unresolved $O(\varepsilon N/(mL))$. Set $N = L\sqrt{m/\varepsilon}$; both become $O(\sqrt{\varepsilon/m})$. □

C Regulatory Infeasibility: Full Derivation

From Theorem 3, verification to accuracy δ requires $m \geq L\varepsilon/\delta^3$. For K demographic groups with minimum proportion $\pi_{\min}$, the effective sample size per group is $m\pi_{\min}$. Each group requires $L\varepsilon/(\pi_{\min}\delta^3)$ total samples. Summing: $m_{\text{total}} \geq K \cdot \frac{L\varepsilon}{\pi_{\min}\delta^3}$.

EU AI Act Annex III (K=10, $\pi_{\min}$=0.05, δ=0.02, ε=0.05, L=1): $m \geq 10 \cdot 1 \cdot 0.05/(0.05 \cdot 0.000008) = 1{,}250{,}000$. Available: ∼707K (CheXpert + MIMIC-CXR + NIH ChestXray14). Gap: 1.8×.

FDA SaMD (K=8, $\pi_{\min}$=0.05, δ=0.01, ε=0.02): $m \geq 3{,}200{,}000$. Gap: 4.5×.

Under active verification, the requirement drops to $K\varepsilon/(\pi_{\min}\delta^2)$, making all three frameworks feasible—but requiring infrastructure that does not yet exist.

Table 5: Per-subject MMLU rankings for the 20 largest subjects. δ_{floor} is the verification floor; gaps below δ_{floor} are noise (marked 55). Instability = fraction of 200 bootstraps where ranking changes.

Subject	n	L405B	L4-Mav	Q3-80B	Max gap	δ_{floor}	Verif.	Instab.
Professional Law	1534	0.72	0.68	0.68	0.040	0.175	55	0.15
Moral Scenarios	895	0.87	0.47	0.67	0.402	0.179	51	0.00
Miscellaneous	783	0.92	0.92	0.94	0.018	0.108	55	0.44
Professional Psychology	612	0.87	0.86	0.87	0.005	0.133	55	0.73
High School Psychology	545	0.93	0.92	0.96	0.037	0.104	55	0.02
High School Macroeconomics	390	0.88	0.78	0.89	0.108	0.137	55	0.27
Elementary Mathematics	378	0.79	0.68	0.89	0.214	0.155	51	0.00
Moral Disputes	346	0.85	0.86	0.85	0.003	0.137	55	0.80
Prehistory	324	0.91	0.90	0.92	0.019	0.116	55	0.46
Philosophy	311	0.86	0.85	0.85	0.016	0.136	55	0.53
High School Biology	310	0.95	0.80	0.95	0.155	0.121	51	0.28
Nutrition	306	0.92	0.87	0.88	0.046	0.126	55	0.43
Professional Accounting	282	0.74	0.70	0.74	0.046	0.168	55	0.48
Professional Medicine	272	0.92	0.88	0.92	0.048	0.118	55	0.47
High School Mathematics	270	0.57	0.07	0.65	0.581	0.215	51	0.00
Clinical Knowledge	265	0.90	0.84	0.90	0.060	0.129	55	0.36
Security Studies	245	0.79	0.81	0.81	0.020	0.151	55	0.55
High School Microeconomics	238	0.95	0.89	0.95	0.063	0.107	55	0.10
High School World History	237	0.92	0.93	0.93	0.008	0.108	55	0.80
Conceptual Physics	235	0.86	0.64	0.90	0.255	0.152	51	0.00

D Extended Benchmark Analysis

Leaderboard noise. Across 57 MMLU subjects, we compute pairwise accuracy gaps between 3 models and compare to per-subject verification floors. Results: 37/57 subjects (65%) are completely unranked; 56/57 (98%) have at least one unranked pair; 132/171 pairwise comparisons (77%) fall below the floor. Mean ranking instability under 80%-bootstrap: 29.1%.

E Full Proofs of Theorems 1 and 3

Theorem 1 (Le Cam). Fix score distribution at $p_0 = 1 - \varepsilon$. P_0: $Y \sim \text{Bern}(1-\varepsilon)$, $\text{ECE} = 0$. P_1: $Y \sim \text{Bern}(1-\varepsilon-\delta)$, $\text{ECE} = \delta$. Both use constant $\Delta \in \mathcal{F}(L)$. $\text{KL}(P_0^m \| P_1^m) = m\delta^2/(\varepsilon(1-\varepsilon)) \le m\delta^2/\varepsilon$. Bretagnolle–Huber: $R^* \ge (\delta/2)e^{-\text{KL}}$. Set KL = 1: $\delta = \sqrt{\varepsilon/m}$, giving $R^* \ge (1/2e)\sqrt{\varepsilon/m}$. □

E.1 Assouad Alternative

Partition $[0,1]$ into B bins. Consider $\Delta_\omega(p) = h\omega_b$ for $p \in I_b$, $\omega \in \{-1,+1\}^B$, $h = L/(2B)$. By Assouad's lemma with $\text{KL} = 4mh^2/(B\varepsilon)$, setting $h = c\sqrt{B\varepsilon/m}$ and $B = (Lm/\varepsilon)^{1/3}$ recovers $R^* \ge c'(L\varepsilon/m)^{1/3}$. □

Theorem 3 (Upper Bound). Bias: L/B. Variance: $\sqrt{\varepsilon B/m}$. Total: $L/B + \sqrt{\varepsilon B/m}$. At $B^* = (L^2 m/\varepsilon)^{1/3}$: $(L\varepsilon/m)^{1/3}$. □

F Compositional Tax Proof and Dynamics

Theorem 12. By the chain rule, $g_K \circ \cdots \circ g_1$ is $(\prod L_k)$-Lipschitz. Since $\Delta_{\text{sys}} = \eta_{\text{sys}} - p$ and $p \mapsto p$ is 1-Lipschitz, $L_{\text{sys}} \le \prod L_k + 1$. Apply Theorem 3 with $L = L_{\text{sys}}$. □

Verification dynamics. Under $\varepsilon(t) = c_0 t^{-\beta}$ and deployment rate r: critical exponent $\beta^* = 1/2$ (passive), $\beta^*_{\text{active}} = 1$ (active). Models improving faster than β^* outrun verification.

Recalibration trap. Iterative recalibration with factor γ: $K^* = \lfloor \log(M_{\text{total}}(1-\gamma)^2 \text{ECE}_0^2/\varepsilon)/(2\log(1/\gamma)) \rfloor$. On MMLU ($M_{\text{total}} = 14{,}000$, $\varepsilon = 0.05$, $\text{ECE}_0 = 0.10$, $\gamma = 0.5$): $K^* = 4$ rounds.

G Full Proof of Theorem 2

Step 1: Reduction to nonparametric regression. Within window $[1-\varepsilon-w/2, 1-\varepsilon+w/2]$ where μ places mass $\geq \rho$, labels are $\text{Bern}(p + \Delta(p))$ with variance $\approx \varepsilon$. The ECE on this window is $\int |\Delta(p)| d\mu$—an L_1 norm.

Step 2: Brown–Low equivalence. Estimating $\int |\Delta| dp$ from m observations with variance ε is equivalent to white noise with $n_{\text{eff}} = m/\varepsilon$ (Brown & Low, 1996).

Step 3: Lepski et al. By Theorem 2.2 of Lepski et al. (1999): $R^* \geq c \cdot (m/\varepsilon)^{-1/3}/(\log m)^{c'}$. Incorporating L: $R^* \geq c_2(L\varepsilon/m)^{1/3}/(\log m)^{c_3}$. □

H White-Box Irrelevance: Full Proof

Fix identical $\mathcal{S}$. Construct P_0 (calibrated) and P_1 (miscalibrated by δ near $p \approx 1-\varepsilon$). $\mathcal{S}$ is identical under both. Conditional KL: $m\delta^2/\varepsilon$, identical to Theorem 1. □

The result extends to any side information independent of fresh labels conditional on scores.

I Sequential Verification

Lower: Wald's identity under Le Cam hypotheses: $\mathbb{E}[\tau] \geq \varepsilon \log(1/\alpha)/\delta^2$. **Upper:** Confidence sequence via method of mixtures (Howard et al., 2021): stop when width $\leq \delta$, giving $\tau \approx \varepsilon \log(1/\alpha)/\delta^2$. Matching.

J Temperature Scaling and Transfer

Proposition (Temperature Scaling). Under correct specification ($\Delta(p;T) = \sigma(\text{logit}(p)/T) - p$, $T^*=1$), MLE achieves $\text{CE} = O_p(1/\sqrt{m})$—the parametric rate. Fisher information: $I(1) = \mathbb{E}[p(1-p)\text{logit}(p)^2] \approx \varepsilon \log^2(1/\varepsilon)$. Misspecification bias persists regardless of m.

Proposition (Verification Transfer). If M_2 differs from verified M_1 by L_h-Lipschitz shift $\leq \delta_{\text{ft}}$: $m_2 = \Theta(L_h \varepsilon_2/\delta_{\text{ft}}^3)$ via transfer vs. $\Theta(L\varepsilon_2/\delta^3)$ from scratch. Saving: $(L/L_h)(\delta/\delta_{\text{ft}})^3$.

Proposition (Ungameability). Constant $\Delta = \delta$ minimizes KL (hardest to detect). Any non-constant Δ with same ECE is easier. $R^* = \Theta(\sqrt{\varepsilon/m})$.

K Corrections and Relaxations

Correlation. If benchmarks share items, effective m is reduced. **Heterogeneity.** Non-uniform score distributions: use equal-mass binning. **Pre-calibration.** Post-hoc calibration changes L and ε; the tax applies to the post-calibration function.

L Numerical Sanity Checks

M Synthetic Experiments

Setup. Calibration gap $\Delta(p) = A\sin(2\pi kp)$ with $L = 2\pi kA$. Scores from $\text{Beta}((1-\varepsilon)/\varepsilon, 1)$. Labels $Y \sim \text{Bern}(p + \Delta(p))$. Optimal B^*.

Results. (1) Phase transition validates at $m\varepsilon \approx 1$ (Fig. 2). (2) Passive slope converges to $-1/3$ as k increases (zero-crossings drive worst-case). (3) Active-passive gap widens with L.

Table 6: Sharp Le Cam constants $c_1(\varepsilon)$ via exact Bernoulli TV.

ε	c_1 (Bretagnolle–Huber)	c_1 (Exact TV)	m_{detect} (δ=0.01)	m_{estimate} (δ=0.01, L=1)
0.01	0.282	0.278	80	10,000
0.05	0.347	0.344	601	50,000
0.10	0.365	0.361	1,333	100,000
0.20	0.371	0.367	2,754	200,000
0.35	0.381	0.377	5,076	350,000

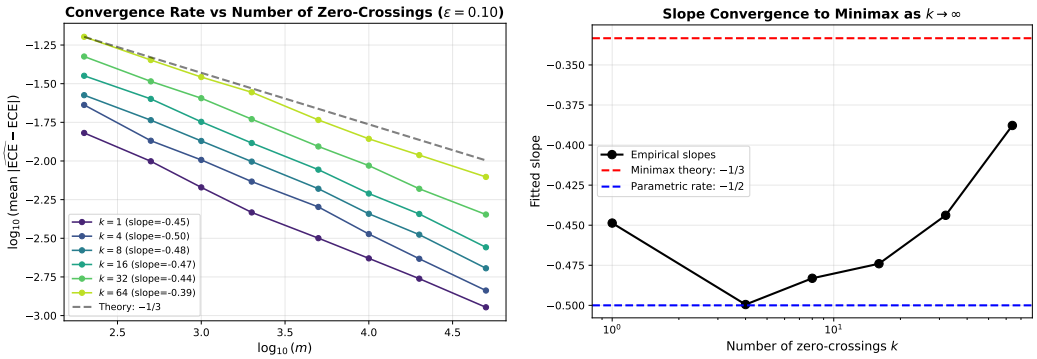

Figure 5: Slope convergence to minimax $-1/3$ as zero-crossings increase (synthetic).

N Real-Model Experiment Details

Three NIM-hosted LLMs on full MMLU test set (N=14,042). Confidence: softmax over A/B/C/D logprobs at first token. Saturation: 78–92% of items yield >0.99 confidence. Lipschitz: 75th-percentile of adjacent-bin slopes (20 bins, min 30 per bin), capped at 5.